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COURSE: Geospatial Information And Science

Semester:2.1

A COMPREHENSIVE LAB REPORT ON THE PROCESS OF DIGITISING A MAP USING QGIS

INTRODUCTION

Geographic Information Systems (GIS) play a critical role in the collection, management, analysis, and presentation of spatial data. One of the fundamental GIS operations is **map digitization**, which involves converting analogue or raster map information into digital vector formats that can be analyzed and visualized within a GIS environment.

QGIS (Quantum Geographic Information System) is a free, open-source GIS software that provides powerful tools for spatial data processing, visualization, and map production. It supports a wide range of raster and vector data formats and allows users to perform georeferencing, digitization, spatial analysis, and cartographic layout creation. Due to its open-source nature, flexibility, and strong community support, QGIS is widely used in academic, research, and professional geospatial applications.

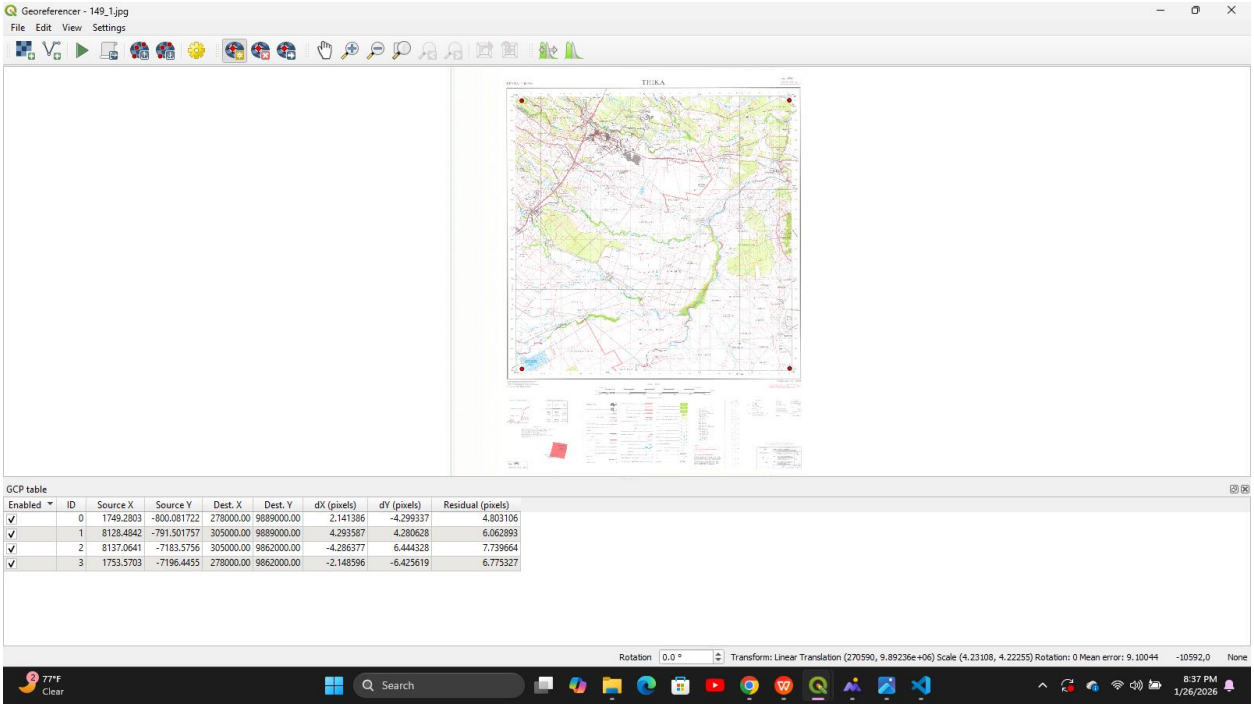
In many regions, including Kenya, historical and hard copy maps exist only in analogue form. To make such maps useful for modern spatial analysis, they must first be **georeferenced** to a known coordinate reference system. In this study, the map was referenced to the **EPSG:21037**- **ARC 1960 Zone 37s datum**, which is the standard geodetic reference system used for mapping in Kenya and much of East Africa.

The aim of this exercise was to use QGIS to georeference a scanned map and digitize key geographical features into separate thematic layers. These layers included roads, railways, schools, towns, and estates. The extracted datasets were then symbolized and organized to produce a clear and accurate map layout suitable for printing and further spatial analysis.

1. Software Preparation and Georeferencing

QGIS was initially closed and restarted to ensure a clean working environment before beginning the digitization process. The scanned map was then loaded into QGIS and georeferenced using the **Georeferencer tool**. The map was referenced to the EPSG:21037-Arc 1960 /UTM Zone 37s **datum**, which is the officially adopted geodetic reference system for Kenya and most East African topographic maps.

Ground Control Points (GCPs) were carefully selected from identifiable and well-distributed locations on the map, such as road intersections and known landmark points. These points were matched with their corresponding real-world coordinates under the ARC 1960 coordinate reference system. The transformation process was successfully completed, producing a georeferenced raster map that aligned correctly with the chosen datum.



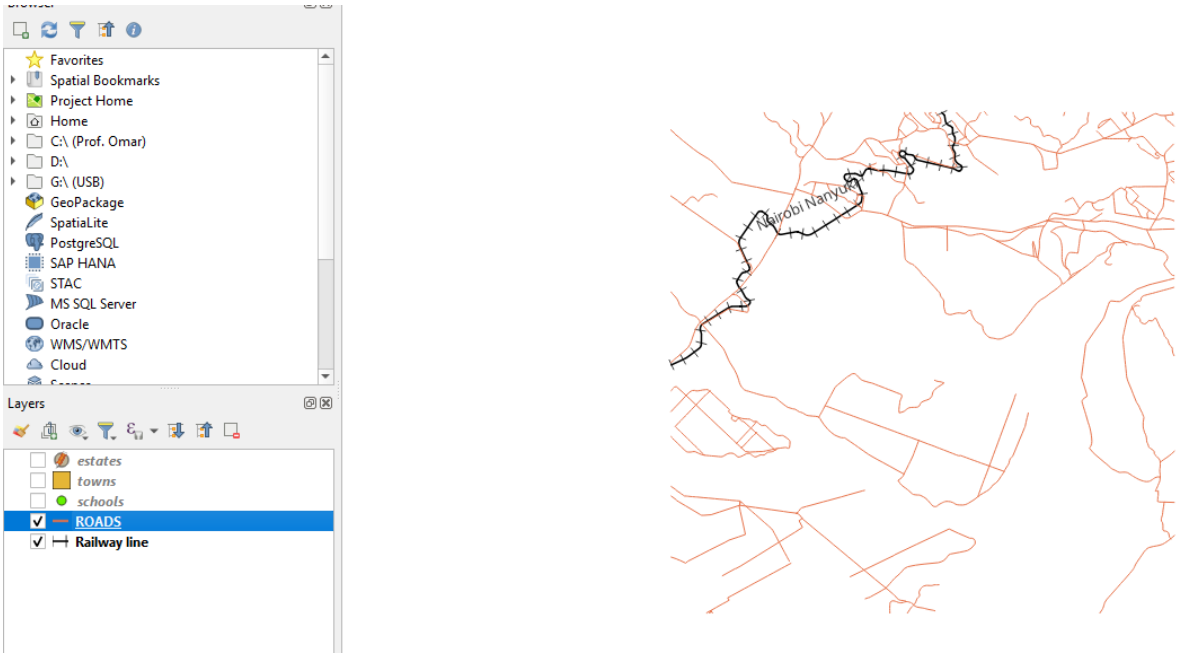
(screenshot1: a figure showing the georeferencing process.)

2. Creation and Digitization of Vector Layers

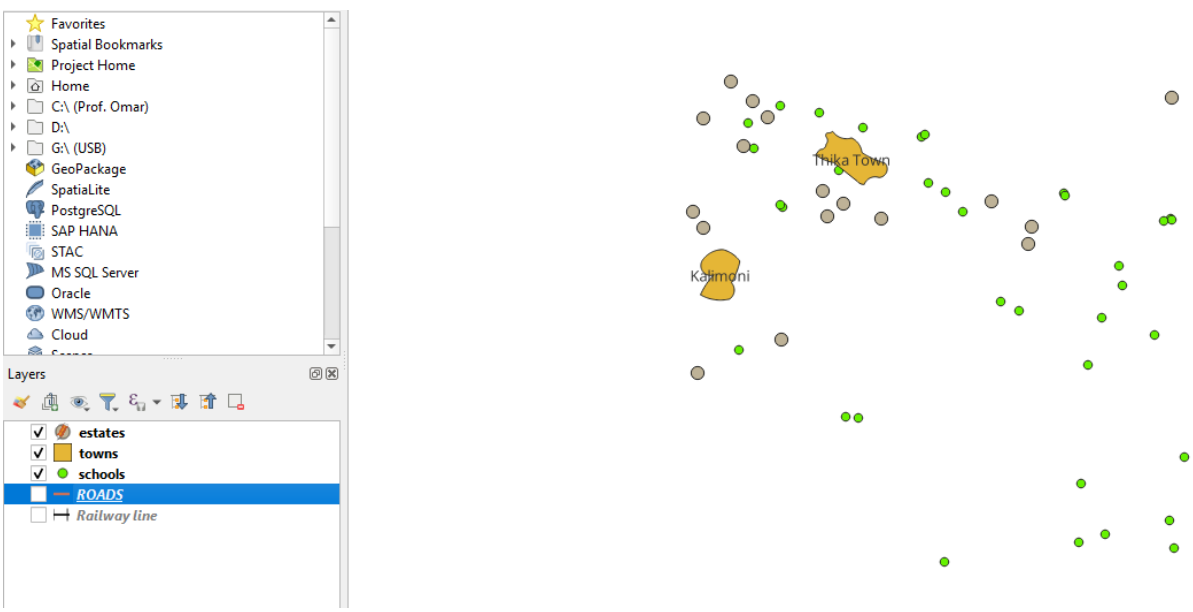
After successful georeferencing, separate vector layers were created to extract individual map features. Each layer was assigned the appropriate geometry type and attribute structure:

- **Roads layer** – line string geometry
- **Railways layer** – line string geometry
- **Schools layer** – point geometry
- **Towns layer** – polygon geometry
- **Estates layer** – point geometry

Digitization was carried out by tracing the respective features directly from the georeferenced map. Care was taken to maintain positional accuracy and logical connectivity, particularly for linear features such as roads and railways. Attribute data such as feature names and types were entered where applicable.



(Screenshot 2: Digitization of road and railway layers)



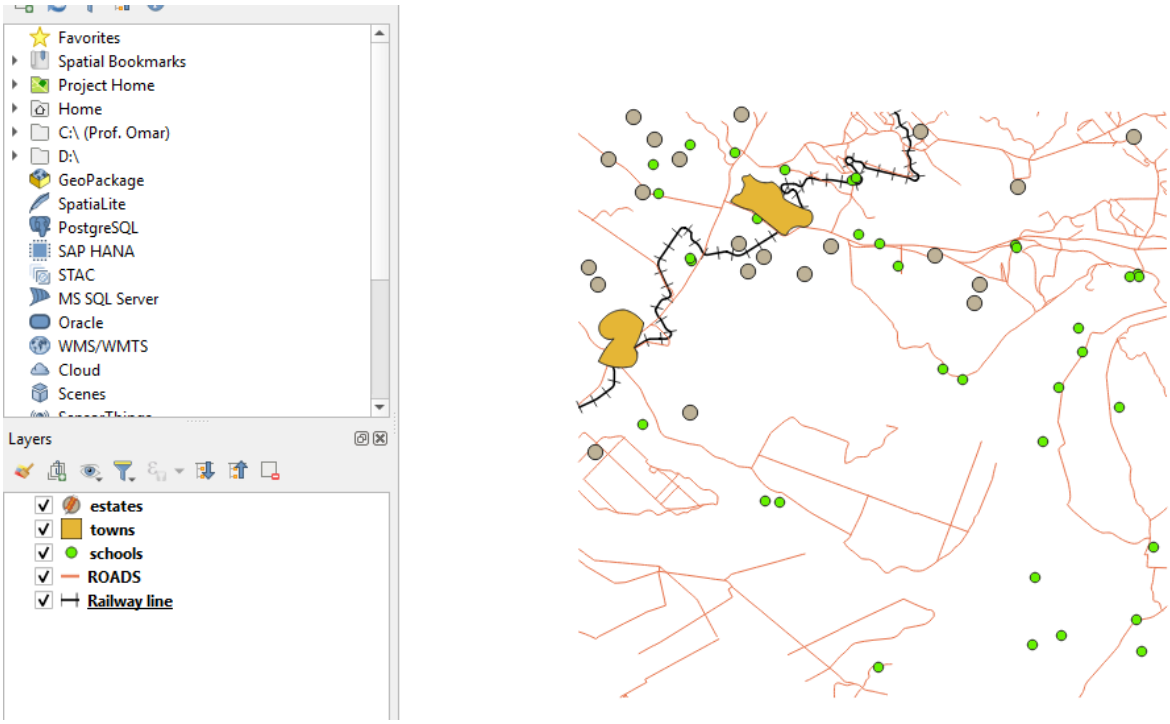
(Screenshot 3: Digitization of schools, towns, and estates layers)

3. Layer Management and Visualization

Each extracted layer was symbolized differently to enhance visual clarity and distinction between feature types. Roads and railways were assigned contrasting line styles, while point features such as schools and towns were symbolized using distinct markers. Estates

were displayed using polygon fills with transparent styling to allow underlying features to remain visible.

The successful overlay of all layers confirmed that the digitized features were spatially consistent and correctly aligned with the referenced map.



(Screenshot 4: QGIS map canvas showing all digitized layers overlaid)

The attributes of the generated vector layers were cleared filled to allow easy access, management and retrieval at will whenever the need to use will arise.

ROADS — Features Total: 104, Filtered: 104, Selected: 0

	id	Name	Type	Width
1	NULL	THIKA ROAD	Murram	12
2	NULL	KANDARA	Murram	6
3	NULL	GARRISSA ROAD	Murram	6
4	NULL	JOSEPH KANGE...	LOOSE SURFACE	6
5	NULL	Kakuzi Road	Loose surface	6
6	NULL	BRIDGES ROAD	Loose surface	6
7	NULL	DONYO SABUK...	Loose surface	6
8	NULL	E496	Murram	6
9	NULL	MAKINDU ROAD	Loose surface	6
10	NULL	MITUBIRI ROAD	Loose surface	6
11	NULL	NULL	Loose surface	6
12	NULL	HARRIES ROAD	Loose surface	6
13	NULL	NULL	Murram road	6
14	NULL	NULL	dry weather road	NULL
15	NULL	NULL	dry weather road	NULL
16	NULL	NULL	dry weather road	NULL
17	NULL	NULL	Dry weather road	NULL
18	NULL	NULL	Dry weather road	NULL
19	NULL	NULL	Dry weather road	NULL
20	NULL	NULL	Dry weather road	NULL
21	NULL	NULL	Dry weather Road	NULL
22	NULL	NULL	Dry weather road	NULL
23	NULL	NULL	Dry Weather road	NULL
24	NULL	NULL	Dry weather road	NULL
25	NULL	NULL	Murrum Road	NULL

(A screenshot showing attribute modification for the extracted layers)

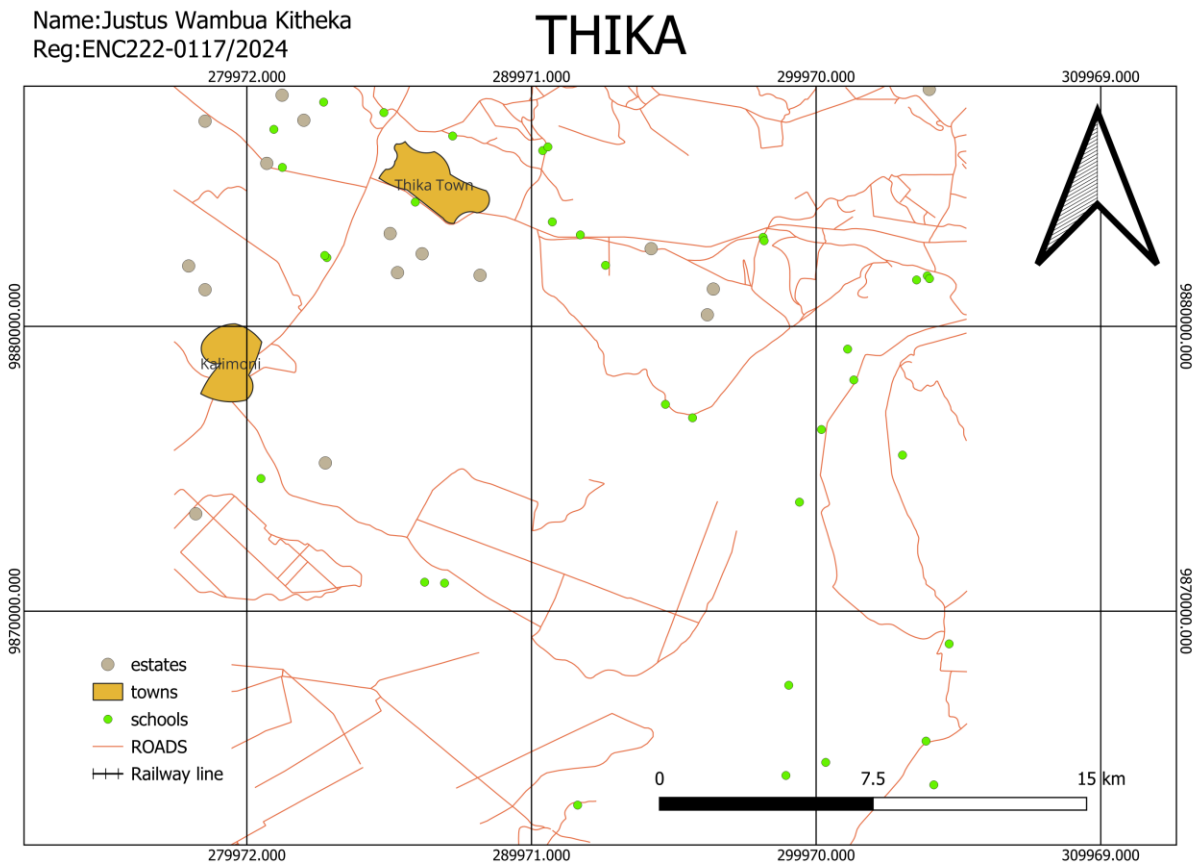
4. Map Layout and Printing

A final map layout was created using the **Print Layout tool** in QGIS. The layout included essential cartographic elements such as:

- Map title
- Legend
- Scale bar

- North arrow
- Coordinate grid
- Source and datum information (ARC 1960)

The layout displayed all digitized layers clearly and was exported in printable format. This final map serves as a visual summary of the digitization results and demonstrates the successful extraction of thematic layers from the original map.



(Image 1: Final printed map layout showing roads, railways, schools, towns, and estates)

5. Overall Outcome.

The digitization process was successfully completed, resulting in accurate and well-structured vector datasets representing key geographic features. Referencing the map to ARC 1960 ensured compatibility with regional geospatial data standards. The extracted layers can now be used for further spatial analysis, mapping, and decision-making purposes.

6. conclusion.

It was concluded that map digitization is an important process of extracting information as well as map analysis. Through this process, required information can be easily retrieved from general maps and can therefore be used for their distinguished purpose.

References

1. https://qgis.org

2. https://qgis.org/

3. https://desktop.arcgis.com/

4. Slocum, T. A., McMaster, R. B., Kessler, F. C., & Howard, H. H. (2009). *Thematic Cartography and Geovisualization* (3rd ed.). Pearson Prentice Hall